

Disproof of Conjecture 00000000405: Spin-Pairing Products of Kostka Numbers Do Not Always Divide $n!$

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Abstract

Conjecture 00000000405 asserts that for every partition $\lambda \vdash n$ the spin-pairing product of Kostka numbers $K_{\lambda, (1^n)} \cdot K_{\lambda', (1^n)}$ divides $n!$. We give an explicit counterexample at the first nontrivial shape: $n = 3$, $\lambda = (2, 1)$, for which the product equals 4 while $3! = 6$, and $4 \nmid 6$.

1 The conjecture

For partitions μ and λ of n , the Kostka number $K_{\lambda\mu}$ counts the semistandard Young tableaux (SSYT) of shape λ and content μ . For $\mu = (1^n)$, the content condition forces each entry $1, \dots, n$ to occur exactly once, so $K_{\lambda, (1^n)} = f^\lambda$, the number of standard Young tableaux (SYT) of shape λ .

Conjecture 00000000405 claims:

$$\forall \lambda \vdash n : \quad K_{\lambda, (1^n)} \cdot K_{\lambda', (1^n)} \mid n!,$$

where λ' denotes the conjugate partition.

2 Counterexample: $n = 3$, $\lambda = (2, 1)$

2.1 Computing $K_{(2,1), (1,1,1)}$

By the hook-length formula for the shape $(2, 1)$, whose hook lengths are 3, 1, 1:

$$f^{(2,1)} = \frac{3!}{3 \cdot 1 \cdot 1} = \frac{6}{3} = 2.$$

The two standard tableaux are

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} \quad \text{and} \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array},$$

and a brute-force enumeration of all 6^3 fillings confirms these are the only SSYT of shape $(2, 1)$ with content $(1, 1, 1)$. Hence $K_{(2,1), (1,1,1)} = 2$.

2.2 Computing $K_{(2,1)', (1,1,1)}$

The shape $(2, 1)$ has two columns, of heights 2 and 1, so its conjugate is $(2, 1)' = (2, 1)$: the partition is self-conjugate. Therefore

$$K_{(2,1)', (1,1,1)} = K_{(2,1), (1,1,1)} = 2.$$

2.3 The divisibility fails

$$K_{(2,1),(1^n)} \cdot K_{(2,1)',(1^n)} = 2 \cdot 2 = 4, \quad 3! = 6, \quad 6 \bmod 4 = 2 \neq 0.$$

Thus $4 \nmid 3!$, and the conjecture is **false**.

3 Boundary of validity

An exhaustive recomputation (see `reproduce.py`) of both Kostka numbers by direct enumeration of SSYT gives:

n	λ	$K_{\lambda,(1^n)}$	$K_{\lambda',(1^n)}$	product	$n!$	divides?
1	(1)	1	1	1	1	yes
2	(2), (1, 1)	1	1	1	2	yes
3	(3), (1, 1, 1)	1	1	1	6	yes
3	(2, 1)	2	2	4	6	no ($6 \bmod 4 = 2$)
4	(2, 2)	2	2	4	24	yes
4	(3, 1), (2, 1, 1)	3	3	9	24	no ($24 \bmod 9 = 6$)
5	(4, 1), (2, 1, 1, 1)	4	4	16	120	no
5	(3, 2), (2, 2, 1)	5	5	25	120	no
5	(3, 1, 1)	6	6	36	120	no

The conjecture holds for $n \leq 2$ (the product is always 1) and for the row and column shapes (n) , (1^n) at every n ; $(2, 1)$ at $n = 3$ is the unique minimal counterexample. Violations are abundant from $n = 4$ onward (65 violating shapes within $n \leq 8$); essentially the conjecture only survives on shapes where f^λ is very small.

4 Reproducibility

- `reproduce.py`: dependency-free Python recomputation by brute-force SSYT enumeration, cross-checked against the hook-length formula.
- `lean4/`: a core-Lean (no Mathlib) encoding; `lake build` followed by `lake env lean Check.lean` certifies every theorem with *zero axioms* (`#print axioms` reports “does not depend on any axioms”), including the brute-force count `sytShape21 = 2`, the self-conjugacy `colHeights [2, 1] = [2, 1]`, and $3! \bmod 4 = 2 \neq 0$.